

Small-cap Biotech Trial Results in 8-K Filings

Companion notebook: `biotech-strategy-notebook.ipynb`

1. Hypothesis

Thesis: A small-cap biotech is usually one or two drug programs, so a clinical trial readout re-prices the whole company in a single session. The result is disclosed in an 8-K tagged `clinical_trial_results`, with the press release attached, and that press release states the endpoints, the effect sizes and the company's next step in language a careful reader can score. The category also lives in exactly the names the market covers least. In 2024–25, eleven of the twelve largest US biopharmas filed no such 8-K at all, and when a big-pharma partner is named on a small biotech's readout, the partner moves about 1.5% that day while the biotech moves about 14%.

We expected two things to be true:

1. The direction is in the text. If a language model can tell a clean win from a miss, and the stock's move by the close of the filing session falls well short of what the text deserves, the market has under-reacted and the move should continue.
2. The hedge is cheapest right after the event. Implied volatility is extreme before a known readout and collapses once the result is out, so the protection a directional readout trade needs can be bought the evening of the 8-K at a discount.

Category-to-strategy link: A positive readout means long stock with a protective put (library strategy 3). A negative readout means short stock with a long call (library strategy 1). Both hedges use a one-month expiry and a strike 10% out-of-the-money, because biotech option chains rarely have liquid longer-dated strikes. The whole rule was written down before any result was looked at.

2. Method: from filing to trade

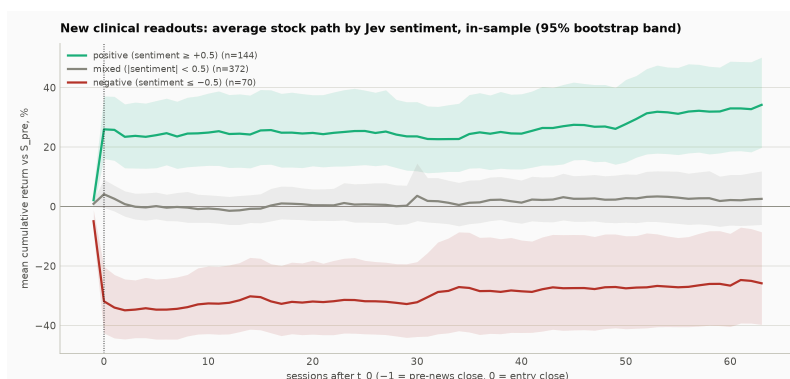
1. **Get Events:** We start from every `clinical_trial_results` 8-K filed by a company in a 1,499-name universe of SIC 2834 and 2836 filers, matched by CIK. The universe includes companies that were later acquired, delisted or went bankrupt, so a failed trial that ends a company shows up as a loss rather than vanishing from the data. Each filing is one event. The in-sample window is 2024–25, with 1,140 filings. The out-of-sample window is January to August 2026, with 333 filings, and nothing was fitted on it.
2. **Check Timing:** The EDGAR acceptance timestamp tells us when the filing actually became public. If it was accepted before 4 pm ET we enter at that day's close, otherwise at the next session's close. Press releases often come out a day before the 8-K, so the model also extracts the announcement date from the text, and the pre-news reference price is taken from the session before that date rather than before the filing. Everything the stock does between that close and our entry is the reaction an 8-K trader has already missed.
3. **Read the Event:** The press release goes to Jev, a language model pinned to one version, with the company name, ticker and drug codes masked so it cannot recognise a famous readout. It answers a fixed set of questions: is this a new readout from a human trial, is it the first disclosure, how positive is the outcome on a 0–4 scale, was the primary endpoint met, is there a safety signal, what does the company do next, which phase, and whether a big-pharma partner is named. To check that it reads correctly, a human labelled 60 filings blind. Jev's outcome agreed within one level 98% of the time, and unmasking the text changed the trade direction in only 1% of filings.
4. **Determine the Signal:** The outcome score becomes a sentiment between -1 and $+1$. The move a readout "deserves" is 0.78 times sentiment times the implied move priced by the one-month straddle before the news, where the 0.78 was fitted in-sample and then frozen. We trade only when the move already realised by entry is less than half of that, which is the under-reaction condition. A filing must also clear a few gates: Jev at least 70% sure it is a new human readout and 60% sure it is the first disclosure, sentiment of at least 0.5 in absolute value, and no serious safety signal on a long.
5. **Trade and Exit:** The long book buys 100 shares and one put; the short book shorts 100 shares and buys one call. Both are entered at the filing-session close against real closing quotes. The position closes at a five-session time stop, fixed in-sample, or earlier if the gap closes or the company files another 8-K. The option is the stop-loss.
6. **Scoring:** Results come in two layers. The signal layer is the bare stock for every event that passes the gates, 94 trades in-sample, and tests whether the text carries information. The trade layer adds the hedge for the events with a priced option chain, 51 trades, and tests whether the strategy is tradeable. Every number is the strategy minus a placebo of the same tickers on random non-event days with random direction, at all eight fixed horizons, with a 95% confidence interval from resampling. Around that sit a random-direction shuffle, a no-model baseline that follows the day-0 move, a 224-cell sensitivity grid, and a cost waterfall built from the actual option quotes.

3. Results

Jev reads the filing correctly, but the market has finished pricing it by the time an 8-K trader can act. The under-reaction trade has no edge. The hedge half of the thesis holds.

Mean P&L per \$1 of stock, strategy minus placebo, by horizon in sessions after entry. * = 95% confidence interval excludes zero.

Layer · window	n	h=1	h=2	h=3	h=5	h=10	h=21	h=42	h=63
Stock only · in-sample 2024–25	94	−0.1%	−0.8%	−1.0%	−1.1%	−3.0%	−5.8%	−9.8%	−7.8%
Stock + hedge · in-sample	51	−1.2%	−1.2%	−0.7%	+1.6%	+1.6%	+0.5%	−10.8%	−7.1%
Stock only · out-of-sample 2026	31	−2.2%*	−3.7%*	−4.6%*	−4.0%	−6.2%	−4.1%	+3.3%	+5.3%
Stock + hedge · out-of-sample	24	−2.6%*	−2.7%*	−3.1%*	−4.6%*	−2.4%	+0.1%	+3.1%	+1.0%



The model's reading lines up with the market level by level. The average day-0 move runs from −32% for a clear failure to +26% for a strongly positive pivotal result, with a rank correlation of 0.38 across 586 new readouts and the right sign 71% of the time. The problem is what happens after entry. The figure shows the average path flat from the entry close onward, and by that close the median readout has only about a quarter of its five-session move left. The reaction is also steepest for the smallest, most concentrated companies, which is exactly the pattern our thesis needed the market to miss. It does not miss it.

The under-reaction rule therefore picks the cases where the market disagreed with the text, and the market was right. The rule sits below the placebo at every in-sample horizon and significantly below it at one to three sessions out-of-sample. When the same events are given a random direction, 77% of the random draws beat the rule in-sample and 94% out-of-sample. A baseline with no model at all, which simply follows the day-0 move, does better, earning about +1.6% in-sample and +2.6% out-of-sample at five sessions. The losses concentrate in the short book, where stocks that have just failed a trial squeeze upward: the in-sample short book is down 49% at 21 sessions on 20 signals.

The option side is a different story. Implied volatility fell in 65% of readouts, by a median of 12%, and the same 10%-out-of-the-money put was 23% cheaper the evening after the 8-K than the evening before in three out of four events. Held to the five-session stop, it cut the long book's worst-decile loss from 16% to 8.5% and the single worst trade from 29% to 10%, for a median strike-implied maximum loss of 18% of the position. Among the library strategies the collar and the protective put rank highest on the long side; nothing rescues the short side.

The null is not an artefact of one setting. It survives sentiment thresholds from 0.25 to 0.75, under-reaction gates from 0.25 to off, a confidence filter, no safety veto, adding `regulatory_decision` filings, strikes 5% to 20% out, expiries from one to six months and borrow fees from 5% to 75%. Two cells deviate, and we flag both as found after the fact. Mid-caps above \$2B are positive at 42 and 63 sessions while micro-caps are strongly negative, and a continuation variant that trades with the day-0 move rather than against it earns 2% to 6% over placebo in-sample and 2% to 4% at short horizons out-of-sample. As a calendar-time portfolio, the hedged book has a Sharpe ratio of 0.11 with a confidence interval from −1.3 to +1.4 and a 30% maximum drawdown.

Trading it in real life: Costs remove what little there is. The median quoted spread on the hedge leg is 53% of its mid, so at the five-session stop a gross +0.9% becomes −3.2% net once the option spread is crossed twice and the stock costs 10 basis points each way; only a quarter of trades finish net positive. Liquidity is thinner still. Of 94 signals, 51 have a listed and traded hedge and 9 pass our pre-registered liquidity gate, and at 10% of day-0 option volume the capacity is about \$550 per trade, under \$2M a year. The option leg, not the stock, is what binds. We had no borrow data for the short side, where small-cap biotech after bad news is hard to borrow and prone to recalls, though the borrow grid barely moves the result because the shorts lose on price, not carry. Entering 30 minutes after acceptance does worse than waiting for the close, so speed is not the missing ingredient. Model risk remains too: Jev is one pinned version, and same-day co-events such as financings are not separated out.

What a portfolio manager can act on: We would not run the under-reaction rule. What does hold is the hedge: if you own a small-cap biotech through a readout, buy protection after the 8-K rather than before. It is about a quarter cheaper, and the 42- and 63-session tails show that the risk is not over once the result is out. The continuation variant and the mid-cap subset are leads for a fresh out-of-sample test, not for capital.

Sealed-window prediction: The notebook prints this before the window is run. The rule shows no edge over placebo at any horizon; the day-0 correlation between sentiment and return replicates at roughly 0.3 to 0.5; the volatility crush and the cheaper post-event put replicate; the continuation variant is the one cell we are unsure of. Counts will be small, about 15 signals a quarter.